WASP-10b

R-1010

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_161231` · created 2026-07-30T02:51:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457753.6682 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.058
Transit depth (Rp/Rs)^2	1.97 +/- 1.61 [%]
Orbital Inclination (inc)	86.62 +/- 1.91
Airmass coefficient 1 (a1)	1.041 +/- 0.072
Airmass coefficient 2 (a2)	-0.035 +/- 0.016
Scatter in the residuals of the lightcurve fit is	6.71 %
Best Comparison Star	#2 - [406, 175]
Optimal Aperture	3.74
Optimal Annulus	13.07
Transit Duration (day)	0.067 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.058 vs 0.1592 (deviation 0.33σ)
Duration fit vs published	0.067 d vs 0.0946 d (deviation 0.95σ)
Mid-time O–C	-7.5 min
Depth SNR	2.4
Residual scatter	6.71 %

Light curve

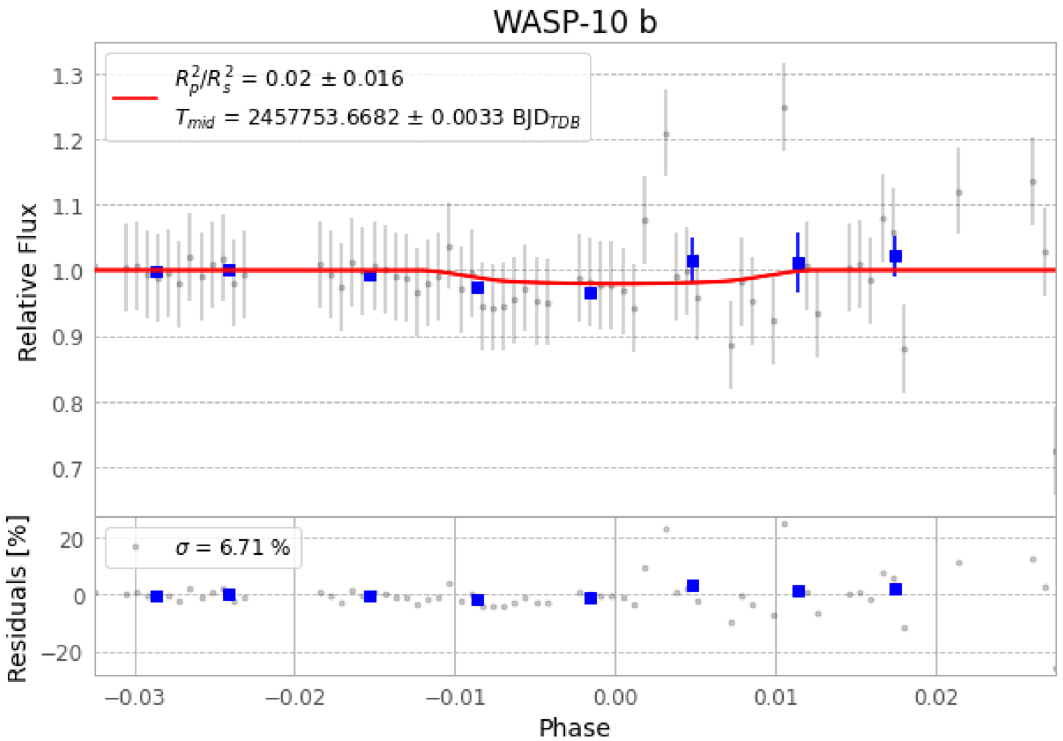

Figure 1 — FinalLightCurve_WASP-10 b_2016-12-30.png (SHA-256 99cf76893bae90ae…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [411, 274], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2e6b88c26056b7f1…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

97 FITS frames (64 MB), `WASP-10161231013643.FITS` → `WASP-10161231063012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-161231.log (8dfd03a2c217…), FinalLightCurve_WASP-10 b_2016-12-30.png (99cf76893bae…), FinalLightCurve_WASP-10 b_2016-12-30.csv (29aa78581955…)

Compute resources

Wall time 280.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 02:51Z.